Data Communication & Computer Networks — Topic 3: Addressing (MAC, IPv4, IPv6, Port, DNS)

Practice Test Booklet — 122 MCQs (2-column, without solutions) · Easy + Medium + Hard mix · Har question ke saath **Source** · Print ke liye · Reference: Forouzan + India-level & worldwide exam papers (GATE/NET/KVS/EMRS/ DSSSB/SSC/RRB/IBPS) · Prepared for: RPSC Programmer • ACP • RSMSSB IA • SA • DD • BCI • SCI • REET • RAS • DSSSB • KVS • NVS • HTET • SSC CGL/CHSL • RRB • IBPS • UPSC & all Rajasthan + Central Govt. Exams (Computer Awareness)

Section A — Address Levels & MAC Address (Physical Address)

Q1. Which of the following is a static hardware address?

निम्न में से कौन-सा static hardware address है?

(A) IP address **(B)** MAC address
(C) Port number **(D)** Domain name

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Easy

Q2. A MAC address is also called a _____ address.

MAC address को _____ address भी कहा जाता है।

(A) Physical address **(B)** Logical address
(C) Source address **(D)** Destination address

[Source: Repo: CN Topic Test-01 (UNIT TEST-1) · also BCI_SCI ST-7] · Medium

Q3. The address used in the Data Link Layer is called:

Data Link Layer में उपयोग होने वाले address को क्या कहते हैं?

(A) Logical address **(B)** Port address
(C) MAC address **(D)** IP address

[Source: Repo: CN Topic Test-04 (UNIT TEST-4)] · Medium

Q4. Which address identifies a unique device on a local network segment?

Local network segment पर कौन-सा address एक unique device की पहचान करता है?

(A) IP address **(B)** MAC address
(C) Gateway **(D)** Subnet mask

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Medium

Q5. What type of MAC address is FF:FF:FF:FF:FF:FF?

FF:FF:FF:FF:FF:FF किस प्रकार का MAC address है?

(A) Unicast **(B)** Multicast
(C) Broadcast **(D)** Loopback

[Source: Repo: CN Topic Test 5 (UNIT TEST-5)] · Medium

Q6. The address used by the data link layer to deliver a frame across one link is called:

Data link layer द्वारा एक link पर frame पहुँचाने के लिए प्रयुक्त address है:

(A) Logical address **(B)** Physical address
(C) Port address **(D)** E-mail address

[Source: Practice — original (textbook-based)] · Medium

Q7. The destination MAC address FF-FF-FF-FF-FF-FF represents:

Destination MAC FF-FF-FF-FF-FF-FF दर्शाता है:

(A) Unicast **(B)** Multicast
(C) Broadcast **(D)** Anycast

[Source: Practice — original (textbook-based)] · Medium

Q8. As a frame moves from hop to hop, which addresses change?

Frame के hop-to-hop जाते समय कौन-से addresses बदलते हैं?

(A) IP addresses **(B)** MAC addresses
(C) Port numbers **(D)** E-mail addresses

[Source: Practice — original (textbook-based)] · Medium

Q9. MAC (hardware) addresses are assigned to NIC manufacturers and administered by:

MAC (hardware) addresses NIC manufacturers को दी जाती हैं और इनका प्रबंधन करता है:

(A) IEEE **(B)** IANA
(C) ICANN **(D)** ITU

[Source: Practice — original (textbook-based)] · Medium

Q10. Which addresses are used in an internet employing the TCP/IP protocols?

TCP/IP protocols वाले internet में कौन-से addresses उपयोग होते हैं?

(A) Physical address and logical address **(B)** Port address
(C) Specific address **(D)** All of these

[Source: LMT MCQ bank (Computer Communication Networks)] · Hard

Q11. What is the size (length) of a MAC address?

MAC address का size (लंबाई) कितना होता है?

(A) 42 bits **(B)** 48 bits
(C) 24 bits **(D)** 36 bits

[Source: RPSC Programmer 2024 Paper-1 (official key) · also BCI_SCI ST-7 / Sanfoundry / InstrumentationTools] · Hard

Q12. An Ethernet destination address 0A-01-12-03-04-05 is a:

Ethernet destination address 0A-01-12-03-04-05 कौन-सा address है?

(A) Unicast address **(B)** Multicast address
(C) Broadcast address **(D)** All of these

[Source: Repo: CN Topic Test-04 (UNIT TEST-4)] · Hard

Q13. Consider the statements: S1: Every computer or Internet device has a unique MAC address. S2: Every computer or Internet device has a unique IP address. Which is correct?

कथन देखिए: S1: हर computer या Internet device का unique MAC address होता है। S2: हर computer या Internet device का unique IP address होता है। कौन-सा सही है?

(A) S1 is true and S2 is false **(B)** Both S1 and S2 are false
(C) S1 is false and S2 is true **(D)** Both S1 and S2 are true

[Source: Repo: CN Topic Test-01 (UNIT TEST-1)] · Hard

Q14. Which header field is rewritten (not removed) when an IP packet passes through a router?

जब एक IP packet router से गुजरता है तो कौन-सा header field बदला (rewrite) जाता है?

(A) Source IP address **(B)** Destination IP address
(C) MAC (Media Access Control) address **(D)** Subnet mask

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Hard

Q15. How many hex digits does a 48-bit MAC address contain?

48-bit MAC address में कितने hex digits होते हैं?

(A) 6 **(B)** 8
(C) 12 **(D)** 16

[Source: Practice — original (textbook-based)] · Hard

Section B — IPv4 Addressing (Format, Classes & Special Addresses)

Q16. The first octet of a Class A IP address ranges from:

Class A IP address का पहला octet किस range में होता है?

(A) 0-127 **(B)** 128-191
(C) 192-223 **(D)** 224-239

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Easy

Q17. Which class of IP address is used for multicasting?

Multicasting के लिए किस class का IP address उपयोग होता है?

(A) Class A **(B)** Class B
(C) Class C **(D)** Class D

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch) · also CN Topic Test-6] · Easy

Q18. Which class of address is reserved for experimental purposes?

प्रायोगिक (experimental) उद्देश्यों के लिए किस class का address reserved है?

(A) Class C **(B)** Class B
(C) Class D **(D)** Class E

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Easy

Q19. The IP address 127.0.0.1 is used for:

IP address 127.0.0.1 किस काम में आती है?

(A) Broadcast **(B)** Class D network
(C) Loopback testing **(D)** APIPA

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch) · also CN Topic Test-03 / CN Topic Test-04] · Easy

Q20. What is the maximum number of unique IPv4 addresses?

Unique IPv4 addresses की अधिकतम संख्या कितनी है?

(A) 2^16 **(B)** 2^24
(C) 2^32 **(D)** 2^64

[Source: IP Addressing quiz (Studocu bank, with keys)] · Medium

Q21. An IPv4 address is divided into how many parts?

IPv4 address कितने भागों में बाँटा जाता है?

(A) 2 **(B)** 3
(C) 4 **(D)** 5

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Medium

Q22. The first octet of a Class B IP address falls between:

Class B IP address का पहला octet किस range में आता है?

(A) 0-127 **(B)** 128-191
(C) 192-223 **(D)** 224-239

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Medium

Q23. How many bits are used for the network portion in a Class C IP address?

Class C IP address में network भाग के लिए कितने bits उपयोग होते हैं?

(A) 8 **(B)** 16
(C) 24 **(D)** 32

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Medium

Q24. Which IP address class provides the largest number of hosts per network?

कौन-सी IP address class प्रति network सबसे अधिक hosts देती है?

(A) Class A **(B)** Class B
(C) Class C **(D)** Class D

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Medium

Q25. Which class of IP addresses supports 65,534 hosts per network?

कौन-सी IP address class प्रति network 65,534 hosts support करती है?

(A) Class A **(B)** Class B
(C) Class C **(D)** Class D

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Medium

Q26. What is the maximum number of IP addresses in a Class C network?

Class C network में IP addresses की अधिकतम संख्या कितनी होती है?

(A) 254 **(B)** 256
(C) 1024 **(D)** 65536

[Source: Repo: CN Topic Test 5 (UNIT TEST-5)] · Medium

Q27. Which of the following is a Class D address?

निम्न में से कौन-सा Class D address है?

(A) 224.1.1.1 **(B)** 233.0.0.10
(C) 239.5.5.5 **(D)** All of the above

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Medium

Q28. Which IP address is reserved for future use and experimental purposes?

कौन-सा IP address भविष्य के उपयोग और प्रायोगिक उद्देश्यों के लिए reserved है?

(A) 127.0.0.1 **(B)** 224.0.0.1
(C) 240.0.0.0 **(D)** 192.168.0.1

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Medium

Q29. Which of the following is a Class B host address?

निम्न में से कौन-सा Class B host address है?

(A) 230.0.0.0 **(B)** 130.4.5.6
(C) 230.7.6.5 **(D)** 30.4.5.6

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Medium

Q30. Which of the following is a valid IPv4 address?

निम्न में से कौन-सा वैध IPv4 address है?

(A) 192.256.1.1 **(B)** 172.31.0.1
(C) 300.100.50.25 **(D)** 500.10.5.2

[Source: 30 important CN MCQs (Medium blog bank)] · Medium

Q31. The address 255.255.255.255 is used for:

Address 255.255.255.255 किसके लिए उपयोग होती है?

(A) A public IP **(B)** A default gateway
(C) A network ID **(D)** Broadcast to all nodes

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Medium

Q32. What is the purpose of the reserved IP address 0.0.0.0?

Reserved IP address 0.0.0.0 का उद्देश्य क्या है?

(A) Default gateway **(B)** Used for DHCP servers
(C) Default route or 'any network' **(D)** Multicast address

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Medium

Q33. The IP address used by a host's packets as the source while the host is being booted (before it gets an IP) is:

जब host boot हो रहा होता है (IP मिलने से पहले) तो उसके packets का source IP address कौन-सा होता है?

(A) 0.0.0.0 **(B)** 1.0.0.0
(C) 1.1.1.1 **(D)** 255.255.255.255

[Source: Repo: CN Topic Test-04 (UNIT TEST-4)] · Medium

Q34. What is the range of addresses for APIPA (Automatic Private IP Addressing)?

APIPA (Automatic Private IP Addressing) की address range क्या है?

(A) 169.254.0.1 to 169.254.255.254 **(B)** 10.0.0.0 to 10.255.255.255
(C) 192.168.0.0 to 192.168.255.255 **(D)** 172.16.0.0 to 172.31.255.255

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Medium

Q35. Which IP address can be used in a WAN (public network)?

कौन-सा IP address WAN (public network) में उपयोग हो सकता है?

(A) 256.0.0.1 **(B)** 172.16.0.10
(C) 15.1.5.6 **(D)** 127.256.0.118

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Medium

Q36. Each computer connected to the internet must:

Internet से जुड़े प्रत्येक computer को क्या होना चाहिए?

(A) Be an IBM PC **(B)** Have a unique IP address
(C) Be internet compatible **(D)** Have a modem connection

[Source: SSC CGL Internet & Networking (via EduRev bank)] · Medium

Q37. Which organization is responsible for managing IP address allocation globally?

वैश्विक स्तर पर IP address allocation का प्रबंधन कौन-सा संगठन करता है?

(A) IEEE **(B)** ICANN
(C) IETF **(D)** ISO

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Medium

Q38. Which of the following is a Class C IP address?

निम्न में से कौन-सा Class C IP address है?

(A) 10.1.1.1 **(B)** 128.5.5.5
(C) 192.168.1.1 **(D)** 225.1.1.1

[Source: Practice — original (textbook-based)] · Medium

Q39. A computer that cannot reach the DHCP server will normally get an address in the range:

जो computer DHCP server तक नहीं पहुँच पाता, उसे किस range का address मिलता है?

(A) 127.x.x.x **(B)** 169.254.x.x
(C) 224.x.x.x **(D)** 0.0.0.0

[Source: Practice — original (textbook-based)] · Medium

Q40. An octet in an IPv4 address can have a value between:

IPv4 address में एक octet का मान हो सकता है:

(A) 0 to 255 / 0 से 255 **(B)** 0 to 128 / 0 से 128
(C) 1 to 256 / 1 से 256 **(D)** 0 to 511 / 0 से 511

[Source: Practice — original (textbook-based)] · Medium

Q41. The IPv4 address block 100.64.0.0/10 is reserved by IANA for:

IPv4 address block 100.64.0.0/10 IANA द्वारा किसके लिए आरक्षित है?

(A) Carrier-grade NAT (shared address space) **(B)** Loopback testing
(C) Multicasting **(D)** Experimental use

[Source: Practice — original (textbook-based)] · Medium

Q42. What is the size of each octet in an IPv4 address?

IPv4 address के प्रत्येक octet का size कितना होता है?

(A) 4 bits **(B)** 8 bits
(C) 16 bits **(D)** 32 bits

[Source: IP Addressing quiz (Studocu bank, with keys)] · Hard

Q43. An IPv4 address is of size:

IPv4 address कितने bits का होता है?

(A) 8 bit **(B)** 16 bit
(C) 32 bit **(D)** 64 bit

[Source: Networking MCQ bank (MQuestions blog)] · Hard

Q44. If the number of networks and number of hosts in Class B are 2^m and (2^n − 2) respectively, then the relation between m and n is:

यदि Class B में networks और hosts की संख्या क्रमशः 2^m और (2^n − 2) है, तो m और n के बीच क्या संबंध है?

(A) 3m = 2n **(B)** 7m = 8n
(C) 8m = 7n **(D)** 2m = 3n

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Hard

Q45. Which subnet mask is the default for the Class B IP address range?

Class B IP address range के लिए default subnet mask कौन-सा है?

(A) 255.255.255.0 **(B)** 255.0.0.0
(C) 255.255.0.0 **(D)** 255.255.255.255

[Source: 30 important CN MCQs (Medium blog bank) · also CN Topic Test-6] · Hard

Q46. What is the primary purpose of NAT (Network Address Translation) in IPv4?

IPv4 में NAT (Network Address Translation) का मुख्य उद्देश्य क्या है?

(A) To map private IP addresses to a public IP address
(B) To increase the security of IP addresses
(C) To facilitate communication between different subnets **(D)** All of the above

[Source: IP Addressing quiz (Studocu bank, with keys)] · Hard

Q47. A host is connected to a department network which is part of a university network; the university network in turn is part of the Internet. Which is the LARGEST network among these?

एक host department network से जुड़ा है जो university network का हिस्सा है; university network बदले में Internet का हिस्सा है। इनमें सबसे बड़ा network कौन-सा है?

(A) The subnet to which the host belongs **(B)** The department network
(C) The university network **(D)** The Internet

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Hard

Q48. Which IP address type is used to communicate with all devices on a subnet?

Subnet के सभी devices से संवाद करने के लिए किस type का IP address उपयोग होता है?

(A) Unicast **(B)** Broadcast
(C) Multicast **(D)** Anycast

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Hard

Q49. Which of the following is NOT a valid private (RFC-1918) IP address?

निम्न में से कौन-सा वैध private (RFC-1918) IP पता नहीं है?

(A) 10.0.0.1 **(B)** 172.16.5.4
(C) 192.168.1.10 **(D)** 11.0.0.1

[Source: CN Topic Test 5.pdf] · Hard

Q50. In the IPv4 addressing format, the number of networks allowed under Class C addresses is:

IPv4 addressing में Class C addresses के अंतर्गत कुल कितने networks allowed हैं?

(A) 2^{14} **(B)** 2^{7}
(C) 2^{21} **(D)** 2^{24}

[Source: GATE CS 2012] · Hard

Q51. Which one of the following CIDR prefixes exactly represents the range of IP addresses 10.12.2.0 to 10.12.3.255?

CIDR prefix में से कौन-सा 10.12.2.0 से 10.12.3.255 तक की IP range को ठीक-ठीक दर्शाता है?

(A) 10.12.2.0/24 **(B)** 10.12.2.0/23
(C) 10.12.0.0/22 **(D)** 10.12.3.0/25

[Source: GATE CS 2024 Set-2] · Hard

Q52. Three machines M, N and P have IP addresses 100.10.5.2, 100.10.5.5 and 100.10.5.6 respectively. The subnet mask is set to 255.255.255.252 for all three machines. Which one of the following is true?

तीन machines M, N, P के IP addresses क्रमशः 100.10.5.2, 100.10.5.5 और 100.10.5.6 हैं। तीनों का subnet mask 255.255.255.252 है। निम्न में से कौन-सा कथन सही है?

(A) All three belong to the same subnet / तीनों एक ही subnet में हैं **(B)** M and N belong to the same subnet / M और N एक ही subnet में हैं
(C) N and P belong to the same subnet / N और P एक ही subnet में हैं **(D)** M and P belong to the same subnet / M और P एक ही subnet में हैं

[Source: GATE CS 2019] · Hard

Q53. Which of the following is NOT a private IP address range?

निम्न में से कौन private IP range नहीं है?

(A) 10.x.x.x **(B)** 172.16.x.x – 172.31.x.x
(C) 192.168.x.x **(D)** 172.32.x.x

[Source: Practice — original (textbook-based)] · Hard

Q54. How many usable host addresses does a Class A network (default mask) have?

Class A network (default mask) में कितने usable host addresses होते हैं?

(A) 2^{24} **(B)** $2^{24} - 2$
(C) $2^{16} - 2$ **(D)** $2^{8} - 2$

[Source: Practice — original (textbook-based)] · Hard

Section C — Subnetting & CIDR

Q55. A subnet /30 is MOST commonly used for:

Subnet /30 सबसे अधिक किसके लिए उपयोग होती है?

(A) Two-host networks **(B)** Four-host networks
(C) Loopback only **(D)** VPN networks

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Medium

Q56. Which part of the IP address 172.16.5.4/24 identifies the network?

IP address 172.16.5.4/24 का कौन-सा भाग network पहचानता है?

(A) 172 **(B)** 172.16
(C) 172.16.5 **(D)** 172.16.5.4

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Medium

Q57. A network ID (address with all host bits = 0) cannot be assigned to:

Network ID (सभी host bits = 0 वाला address) किसे assign नहीं किया जा सकता?

(A) A router **(B)** A switch
(C) A host device **(D)** The entire network

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Medium

Q58. A host has IP 192.168.10.45/27. Which statement is TRUE?

एक host का IP 192.168.10.45/27 है। कौन-सा कथन सही है?

(A) Broadcast address = 192.168.10.63 **(B)** Network ID = 192.168.10.32
(C) First usable host = 192.168.10.33 **(D)** All of the above

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch) · also CN Topic Test-03] · Medium

Q59. In NAT overload (PAT), many private hosts share:

NAT overload (PAT) में कई private hosts साझा करते हैं:

(A) One public IP with different port numbers / एक public IP, अलग-अलग ports के साथ **(B)** One private IP / एक private IP
(C) Different public IPs / अलग-अलग public IPs **(D)** No IP at all / कोई IP नहीं

[Source: Practice — original (textbook-based)] · Medium

Q60. Route aggregation (summarization) is used to:

Route aggregation का प्रयोग होता है:

(A) Increase security / Security बढ़ाने के लिए **(B)** Reduce the size of routing tables / Routing tables छोटी करने के लिए
(C) Increase speed of cables / Cables की speed बढ़ाने के लिए
(D) Encrypt the packets / Packets encrypt करने के लिए

[Source: Practice — original (textbook-based)] · Medium

Q61. For point-to-point links between routers, a /30 prefix is preferred because it provides exactly _____ usable host addresses:

Routers के बीच point-to-point links के लिए /30 prefix क्यों पसंद किया जाता है? यह ठीक _____ usable addresses देता है:

(A) 2 **(B)** 4
(C) 6 **(D)** 14

[Source: Practice — original (textbook-based)] · Medium

Q62. What does the term 'subnetting' refer to in networking?

Networking में 'subnetting' शब्द का क्या अर्थ है?

(A) Dividing a large network into smaller logical networks **(B)** Allocating IP addresses to devices
(C) Filtering network traffic **(D)** Assigning MAC addresses

[Source: 30 important CN MCQs (Medium blog bank) · also CN Topic Test-03] · Hard

Q63. When IPv4 addressing is manually configured, which property identifies the network and host portions of the address?

जब IPv4 addressing manually configure होती है, तो कौन-सा property address के network और host भाग पहचानती है?

(A) DNS server address **(B)** Subnet mask
(C) Default gateway **(D)** DHCP server address

[Source: IP Addressing quiz (Studocu bank, with keys)] · Hard

Q64. In a subnet mask, the number of 1's represents:

Subnet mask में 1's की संख्या किसका प्रतिनिधित्व करती है?

(A) Host bits **(B)** Network bits
(C) Broadcast bits **(D)** Subnet bits

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Hard

Q65. In CIDR notation, what does /16 represent?

CIDR notation में /16 का क्या अर्थ है?

(A) 16 host bits **(B)** 16 network bits
(C) 16 bytes **(D)** 16 subnets

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Hard

Q66. What does /24 denote in CIDR?

CIDR में /24 किसे दर्शाता है?

(A) 24 bits are used for the host portion **(B)** 24 bits are used for the network portion
(C) 24 subnets are created **(D)** 24 addresses are available in the subnet

[Source: RPSC Programmer 2024 Paper-1 (official key)] · Hard

Q67. How many USABLE hosts are there in a /24 subnet?

/24 subnet में कितने उपयोगी (usable) hosts होते हैं?

(A) 128 **(B)** 256
(C) 254 **(D)** 512

[Source: Repo: CN Topic Test 5 (UNIT TEST-5) · also CN Topic Test-03] · Hard

Q68. How many usable hosts are there in a /30 subnet?

/30 subnet में कितने उपयोगी hosts होते हैं?

(A) 2 **(B)** 4
(C) 6 **(D)** 14

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Hard

Q69. A subnet mask of 255.255.255.248 allows how many usable hosts?

Subnet mask 255.255.255.248 कितने उपयोगी hosts की अनुमति देता है?

(A) 2 **(B)** 6
(C) 14 **(D)** 30

[Source: Repo: CN Topic Test 5 (UNIT TEST-5)] · Hard

Q70. The subnet mask 255.255.255.128 provides how many subnets for a Class C network?

Subnet mask 255.255.255.128, Class C network के लिए कितने subnets देता है?

(A) 1 **(B)** 2
(C) 4 **(D)** 8

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Hard

Q71. A network needs 5 subnets, each with at least 16 hosts. Which subnet mask (Class C) should be used?

एक network को 5 subnets चाहिए, हर एक में कम से कम 16 hosts। कौन-सा subnet mask (Class C) उपयोग करें?

(A) 255.255.255.192 **(B)** 255.255.255.248
(C) 255.255.255.240 **(D)** 255.255.255.224

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Hard

Q72. A subnet mask for Class C can have how many 1's by default (remaining bits 0)?

Class C के लिए subnet mask में डिफ़ॉल्ट रूप से कितने 1's हो सकते हैं (बाकी bits 0)?

(A) 10 **(B)** 24
(C) 12 **(D)** 7

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Hard

Q73. What is the valid host range for the subnet 192.168.1.0/25?

Subnet 192.168.1.0/25 के लिए वैध host range क्या है?

(A) 192.168.1.0 - 192.168.1.255 **(B)** 192.168.1.1 - 192.168.1.126
(C) 192.168.1.1 - 192.168.1.254 **(D)** 192.168.1.0 - 192.168.1.127

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Hard

Q74. In subnetting, reducing the number of host bits does what?

Subnetting में host bits की संख्या घटाने से क्या होता है?

(A) Increases the broadcast domain **(B)** Decreases the number of subnets
(C) Increases the number of subnets **(D)** Increases the number of hosts

[Source: Repo: CN Topic Test 5 (UNIT TEST-5)] · Hard

Q75. In a class B subnet: IP address 125.134.112.66, mask 255.255.224.0. What is the first address (network address)?

Class B subnet में: IP address 125.134.112.66, mask 255.255.224.0। पहला address (network address) क्या होगा?

(A) 125.134.96.0 **(B)** 125.134.112.0
(C) 125.134.112.66 **(D)** 125.134.0.0

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Hard

Q76. A Class B network is to be split into subnets with a 3-bit subnet number. What is the maximum number of subnets and hosts in each subnet?

एक Class B network को 3-bit subnet number में बाँटना है। प्रत्येक subnet में subnets और hosts की अधिकतम संख्या कितनी होगी?

(A) 8 subnets and 262141 hosts **(B)** 6 subnets and 262141 hosts
(C) 6 subnets and 1022 hosts **(D)** None of the above

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Hard

Q77. IP address 172.16.123.213 has subnet mask 255.255.255.240. What is its network (first) address?

IP address 172.16.123.213 का subnet mask 255.255.255.240 है। इसका network (पहला) address क्या है?

(A) 172.16.123.240 **(B)** 172.16.123.208
(C) 172.16.123.213 **(D)** 172.16.123.245

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Hard

Q78. An organization has a Class B network and wishes to form subnets for 64 departments. The subnet mask would be:

एक संगठन के पास Class B network है और वह 64 departments के लिए subnets बनाना चाहता है। Subnet mask क्या होगा?

(A) 255.255.128.0 **(B)** 255.255.0.0
(C) 255.255.64.0 **(D)** 255.255.252.0

[Source: RPSC Programmer 2024 Paper-1 (official key)] · Hard

Q79. A Class B network has a subnet mask of 255.255.248.0. What is the maximum number of hosts per subnet?

एक Class B network का subnet mask 255.255.248.0 है। प्रति subnet hosts की अधिकतम संख्या कितनी है?

(A) 2046 **(B)** 1022
(C) 1023 **(D)** 1024

[Source: RPSC Programmer 2024 Paper-1 (official key) · Rajasthan PYQ Q56] · Hard

Q80. An IP address 156.233.42.56 is configured with a 7-bit subnet mask (7 bits borrowed from the host part of the classful network). How many hosts and subnets are possible?

IP address 156.233.42.56 को 7-bit subnet mask (classful network के host भाग से 7 bits उधार) के साथ configure किया गया है। कितने hosts और subnets संभव हैं?

(A) 126 hosts, 510 subnets **(B)** 128 hosts, 512 subnets
(C) 510 hosts, 126 subnets **(D)** 512 hosts, 128 subnets

[Source: Unit Test -2 Solutions (OCR), p5] · Hard

Q81. If a Class B network on the internet has a subnet mask of 255.255.248.0, what is the maximum number of hosts per subnet?

यदि internet पर किसी Class B network की subnet mask 255.255.248.0 हो, तो हर subnet में अधिकतम hosts कितने होंगे?

(A) 1023 **(B)** 1022
(C) 1024 **(D)** 2046

[Source: RPSC Programmer 2024 Paper-1 (via Suraku Academy official-key solutions)] · Hard

Q82. The address of a Class B host is to be split into subnets with a 6-bit subnet number. What is the maximum number of subnets and the maximum number of hosts in each subnet?

Class B host के पते को 6-bit subnet number से subnets में बाँटना है. अधिकतम subnets और हर subnet में अधिकतम hosts कितने होंगे?

(A) 62 subnets and 262142 hosts **(B)** 64 subnets and 262142 hosts
(C) 62 subnets and 1022 hosts **(D)** 64 subnets and 1024 hosts

[Source: RPSC Programmer 2011 Paper-1 (via pyqp.in) + GATE CS 2005] · Hard

Q83. Suppose that in an IP-over-Ethernet network, a machine X wishes to find the MAC address of another machine Y in its subnet. Which one of the following techniques can be used for this?

IP-over-Ethernet network में machine X को अपने ही subnet की machine Y का MAC address जानना है। इसके लिए निम्न में से कौन-सी technique प्रयोग होगी?

(A) ARP request broadcast **(B)** RARP request broadcast
(C) DNS query **(D)** ICMP query

[Source: GATE CS 2019] · Hard

Q84. Node X has a TCP connection open to node Y. Packets from X to Y go through an intermediate IP router R; Ethernet switch S is the first switch on the path. For a packet sent from X, which of the following is TRUE about its destination IP and MAC addresses at the time it leaves X?

Node X की node Y से TCP connection है; packets router R से होकर जाते हैं और पहला Ethernet switch S है। X से निकलते packet के destination IP और MAC addresses के बारे में क्या सही है?

(A) Destination IP = Y's IP; destination MAC = R's MAC / Destination IP = Y का IP; destination MAC = R का MAC **(B)** Destination IP = R's IP; destination MAC = R's MAC / Destination IP = R का IP; destination MAC = R का MAC
(C) Destination IP = Y's IP; destination MAC = Y's MAC / Destination IP = Y का IP; destination MAC = Y का MAC **(D)** Destination IP = R's IP; destination MAC = S's MAC / Destination IP = R का IP; destination MAC = S का MAC

[Source: GATE CS 2025 Set-2] · Hard

Q85. With a /26 subnet mask, how many usable host addresses are there per subnet?

/26 subnet mask पर हर subnet में कितने usable hosts होंगे?

(A) 30 **(B)** 62
(C) 64 **(D)** 126

[Source: Practice — original (textbook-based)] · Hard

Q86. The subnet mask 255.255.255.224 divides a /24 network into subnets of size:

Mask 255.255.255.224, /24 network को कितने size के subnets में बाँटता है?

(A) 32 addresses **(B)** 64 addresses
(C) 16 addresses **(D)** 8 addresses

[Source: Practice — original (textbook-based)] · Hard

Q87. A network 192.168.10.0/24 is divided using /27. How many subnets are created?

192.168.10.0/24 को /27 से बाँटा गया। कितने subnets बनेंगे?

(A) 4 **(B)** 8
(C) 16 **(D)** 32

[Source: Practice — original (textbook-based)] · Hard

Q88. In VLSM, different subnets of the same network use:

VLSM में एक ही network के अलग-अलग subnets प्रयोग करते हैं:

(A) The same mask everywhere / सब जगह एक ही mask
(B) Masks of different lengths / अलग-अलग लंबाई के masks
(C) No mask / कोई mask नहीं **(D)** Only /24 / केवल /24

[Source: Practice — original (textbook-based)] · Hard

Q89. The usable host range of the CIDR block 172.16.45.64/26 is:

CIDR block 172.16.45.64/26 की usable host range है:

(A) 172.16.45.65 – 172.16.45.126 **(B)** 172.16.45.65 – 172.16.45.127
(C) 172.16.45.64 – 172.16.45.127 **(D)** 172.16.45.1 – 172.16.45.254

[Source: Practice — original (textbook-based)] · Hard

Q90. An organization has the block 192.168.10.0/24 and needs subnets for 50, 25 and 10 hosts. Which set of prefix lengths wastes the LEAST address space?

एक संस्था के पास block 192.168.10.0/24 है और उसे 50, 25 और 10 hosts के लिए subnets चाहिए। कौन-सी prefix lengths की जोड़ी सबसे कम address space व्यर्थ करेगी?

(A) /26, /27 and /28 / /26, /27 और /28 **(B)** /26, /26 and /26 / /26, /26 और /26
(C) /27, /27 and /27 / /27, /27 और /27 **(D)** /25, /26 and /27 / /25, /26 और /27

[Source: Practice — original (RPSC Programmer/ACP/DD exam-pattern, textbook-based)] · Hard

Q91. The Cisco IOS command 'ip classless' makes a router:

Cisco IOS का आदेश 'ip classless' router को क्या बनाता है?

(A) Forward packets to the default route even when the destination subnet is not in the routing table / तब भी default route पर packets भेजे जब destination subnet routing table में न हो **(B)** Drop all packets for classful networks / Classful networks के सभी packets गिराए
(C) Advertise only classful routes / सिर्फ classful routes advertise करे **(D)** Disable subnetting on all interfaces / सभी interfaces पर subnetting बंद करे

[Source: Practice — original (RPSC Programmer/ACP/DD exam-pattern, textbook-based)] · Hard

Section D — IPv6 Addressing

Q92. Which IPv6 address type is globally unique and globally routable?

कौन-सा IPv6 address type globally unique और globally routable होता है?

(A) Link-Local **(B)** Multicast
(C) Unique Local **(D)** Global Unicast

[Source: Repo: CN Topic Test 5 (UNIT TEST-5)] · Medium

Q93. IPv6 was developed by:

IPv6 किसने develop किया?

(A) IETF **(B)** ANSI
(C) ISO **(D)** IEEE

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Medium

Q94. Two IPv6 nodes must interoperate but are connected through intervening IPv4 routers. The best solution is:

दो IPv6 nodes को interoperable होना है लेकिन वे बीच में आने वाले IPv4 routers से जुड़े हैं। सबसे अच्छा समाधान क्या है?

(A) Use the dual-stack approach **(B)** Tunneling
(C) No solution **(D)** Replace the system

[Source: Sanfoundry CN MCQs — IPv6] · Medium

Q95. Which DNS resource record should be queried to retrieve the IPv6 address associated with a name?

किसी name से जुड़ा IPv6 address पाने के लिए कौन-सी DNS resource record query करनी चाहिए?

(A) The AAAA record **(B)** The A record
(C) The NS record **(D)** The MX record

[Source: cnp3 MCQ (Bonaventure CN textbook), DNS1] · Medium

Q96. IPv6 addresses are _____ bit addresses.

IPv6 addresses _____ bit के पते होते हैं.

(A) 32 **(B)** 64
(C) 16 **(D)** 128

[Source: SBI/IBPS Computer Awareness Capsule 2018 (IBT India)] · Medium

Q97. In IPv6, a host can generate its own address automatically without any DHCP server using:

IPv6 में host बिना DHCP server के अपना address स्वयं बना सकता है — किसके द्वारा?

(A) SLAAC (Stateless Address Autoconfiguration) **(B)** ARP
(C) BOOTP **(D)** NAT-PT

[Source: Practice — original (textbook-based)] · Medium

Q98. The prefix used for IPv6 link-local addresses (usable only on the local link) is:

IPv6 link-local addresses (सिर्फ उसी link पर काम करने वाले) का prefix होता है:

(A) FE80::/10 **(B)** 2000::/3
(C) FF00::/8 **(D)** ::1/128

[Source: Practice — original (textbook-based)] · Medium

Q99. In IPv6, how many bits are in the address?

IPv6 में address कितने bits का होता है?

(A) 32 **(B)** 64
(C) 128 **(D)** 256

[Source: Repo: CN Topic Test-6 (UNIT TEST-6) · Sanfoundry / Medium / MQuestions banks] · Hard

Q100. IPv6 does NOT use which type of address?

IPv6 किस type का address उपयोग नहीं करता?

(A) Broadcast **(B)** Multicast
(C) Anycast **(D)** None of the mentioned

[Source: Sanfoundry CN MCQs — IPv6] · Hard

Q101. Which features are present in IPv4 but NOT in IPv6?

कौन-सी features IPv4 में हैं लेकिन IPv6 में नहीं?

(A) Fragmentation by routers **(B)** Header checksum
(C) Options **(D)** All of the mentioned

[Source: Sanfoundry CN MCQs — IPv6] · Hard

Q102. Which of the following is TRUE for IPv6?

IPv6 के लिए निम्न में से क्या सही है?

(A) Uses 32-bit addresses **(B)** Includes a header checksum
(C) Supports anycast addressing **(D)** Does not support multicast

[Source: Repo: CN Topic Test 5 (UNIT TEST-5)] · Hard

Q103. Which technique carries an IPv6 packet inside an IPv4 packet?

कौन-सी technique IPv6 packet को IPv4 के अंदर ले जाती है?

(A) Dual stack **(B)** Tunneling
(C) Fragmentation **(D)** NAT

[Source: Practice — original (textbook-based)] · Hard

Q104. Which statement about the IPv6 header is TRUE?

IPv6 header के बारे में कौन-सा कथन सही है?

(A) It has a fixed size of 40 bytes / इसका size 40 bytes fixed है **(B)** It is variable from 20 to 60 bytes / यह 20 से 60 bytes तक बदलता है
(C) It contains a header checksum / इसमें header checksum होता है **(D)** It has a broadcast flag / इसमें broadcast flag होता है

[Source: Practice — original (textbook-based)] · Hard

Section E — Port Numbers, DNS/URL & Address Resolution (ARP)

Q105. Which of the following sets of port numbers belongs to the 'well-known' ports category?

निम्न में से कौन-से port numbers 'well-known' ports category में आते हैं?

(A) 80, 443, 8080 **(B)** 49152, 110, 53
(C) 22, 25, 80 **(D)** 1024, 3306, 8080

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Easy

Q106. What is the typical range of ephemeral (dynamic) ports?

Ephemeral (dynamic) ports की सामान्य range क्या है?

(A) 0-1023 **(B)** 1024-49151
(C) 49152-65535 **(D)** 100-200

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch) · also CN Topic Test 5] · Easy

Q107. The transport layer uses which type of addressing?

Transport layer किस प्रकार की addressing उपयोग करता है?

(A) IP address **(B)** MAC address
(C) Port number **(D)** Domain name

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Easy

Q108. A URL (like www.example.com) belongs to which layer?

URL (जैसे www.example.com) किस layer से संबंधित है?

(A) Presentation **(B)** Application
(C) Transport **(D)** Network

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Easy

Q109. DNS works at which layer of the OSI model?

DNS, OSI model की किस layer पर काम करता है?

(A) Transport **(B)** Application
(C) Network **(D)** Data link

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Easy

Q110. Which of the following uniquely identifies a process on a host?

Host पर किसी process की unique पहचान कौन करता है?

(A) IP address only **(B)** Port number only
(C) IP address + Port number **(D)** MAC address

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)] · Medium

Q111. What is the default port number for DNS?

DNS का default port number क्या है?

(A) 80 **(B)** 21
(C) 53 **(D)** 110

[Source: Repo: CN Topic Test 6 (UNIT TEST-6)] · Medium

Q112. DNS is the abbreviation of:

DNS का पूरा नाम (full form) क्या है?

(A) Dynamic Name System **(B)** Dynamic Network System
(C) Domain Name System **(D)** Domain Network Service

[Source: Networking MCQ bank (MQuestions blog)] · Medium

Q113. Which DNS record type maps a domain name to an IPv4 address?

कौन-सी DNS record type domain name को IPv4 address से जोड़ती है?

(A) CNAME **(B)** A
(C) MX **(D)** PTR

[Source: Repo: CN Topic Test 6 (UNIT TEST-6) · also cnp3 DNS2] · Medium

Q114. What is the range of well-known port numbers?

Well-known port numbers की range है:

(A) 0 – 1023 **(B)** 1024 – 49151
(C) 49152 – 65535 **(D)** 0 – 65535

[Source: Practice — original (textbook-based)] · Medium

Q115. Which DNS record type is used to find the domain name from an IP address?

IP से domain name जानने के लिए कौन-सा DNS record प्रयोग होता है?

(A) A **(B)** MX
(C) PTR **(D)** CNAME

[Source: Practice — original (textbook-based)] · Medium

Q116. A socket address is the combination of:

Socket address किसका संयोजन है?

(A) MAC address and IP address / MAC और IP address
(B) IP address and port number / IP address और port number
(C) MAC address and port number / MAC और port number
(D) URL and e-mail address / URL और e-mail address

[Source: Practice — original (textbook-based)] · Medium

Q117. DNS zone transfers between primary and secondary name servers use:

Primary और secondary name servers के बीच DNS zone transfer प्रयोग करती है:

(A) TCP port 53 **(B)** UDP port 53
(C) UDP port 69 **(D)** TCP port 25

[Source: Practice — original (textbook-based)] · Medium

Q118. What is the role of DNS? It is best described as:

DNS की भूमिका क्या है? सबसे सही विवरण है:

(A) Maps a domain name to an IP address **(B)** Translates a host's private IP to a public IP
(C) Translates a host's IPv6 address to an IPv4 address
(D) Maps a MAC address to an IP address

[Source: RPSC Programmer 2024 Paper-1 (official key) · also CN Topic Test-02 / Medium bank] · Hard

Q119. A reverse lookup in DNS means a lookup to _____ using _____ records.

DNS में reverse lookup का अर्थ है _____ records का उपयोग करके _____ की lookup।

(A) Identify mail servers, SOA **(B)** Map an IP address to a domain name, PTR
(C) Map domain names to IP addresses, A **(D)** Determine the authoritative name server, NS

[Source: RPSC Programmer 2024 Paper-1 (official key) · also CN Topic Test 5] · Hard

Q120. The Address Resolution Protocol (ARP) is used for:

Address Resolution Protocol (ARP) किस काम में आता है?

(A) Finding the IP address that corresponds to a MAC address **(B)** Finding the MAC address that corresponds to an IP address
(C) Finding the IP address from the DNS name **(D)** Finding the IP address of the default gateway

[Source: RPSC Programmer 2024 Paper-1 (official key)] · Hard

Q121. Which one of the following protocols is NOT used to resolve one form of address into another?

निम्न में से कौन-सा protocol एक प्रकार के पते को दूसरे में बदलने के लिए उपयोग नहीं होता?

(A) DNS **(B)** ARP
(C) DHCP **(D)** RARP

[Source: GATE CS PYQ (via Computer Science Junction / examprev)] · Hard

Q122. A reverse lookup in DNS means a lookup to _____ using _____ records.

DNS में reverse lookup का अर्थ है _____ records का उपयोग करके _____ की खोज.

(A) map domain names to IP address, A **(B)** map IP address to domain name, PTR
(C) determine the authoritative name servers, CNAME
(D) identify mail servers, SOA

[Source: RPSC Programmer 2024 Paper-1 (via Suraku Academy official-key solutions)] · Hard

Answer Key — sirf letters (no solutions)

Q1: B **Q2:** A **Q3:** C **Q4:** B **Q5:** C **Q6:** B **Q7:** C **Q8:** B **Q9:** A **Q10:** D
Q11: B **Q12:** A **Q13:** A **Q14:** C **Q15:** C **Q16:** A **Q17:** D **Q18:** D **Q19:** C **Q20:** C
Q21: A **Q22:** B **Q23:** C **Q24:** A **Q25:** B **Q26:** B **Q27:** D **Q28:** C **Q29:** B **Q30:** B
Q31: D **Q32:** C **Q33:** A **Q34:** A **Q35:** C **Q36:** B **Q37:** B **Q38:** C **Q39:** B **Q40:** A
Q41: A **Q42:** B **Q43:** C **Q44:** C **Q45:** C **Q46:** A **Q47:** D **Q48:** B **Q49:** D **Q50:** C
Q51: B **Q52:** C **Q53:** D **Q54:** B **Q55:** A **Q56:** C **Q57:** C **Q58:** D **Q59:** A **Q60:** B
Q61: A **Q62:** A **Q63:** B **Q64:** B **Q65:** B **Q66:** B **Q67:** C **Q68:** A **Q69:** B **Q70:** B
Q71: D **Q72:** B **Q73:** B **Q74:** C **Q75:** A **Q76:** D **Q77:** B **Q78:** D **Q79:** A **Q80:** A
Q81: D **Q82:** C **Q83:** A **Q84:** A **Q85:** B **Q86:** A **Q87:** B **Q88:** B **Q89:** A **Q90:** A
Q91: A **Q92:** D **Q93:** A **Q94:** B **Q95:** A **Q96:** D **Q97:** A **Q98:** A **Q99:** C **Q100:** A
Q101: D **Q102:** C **Q103:** B **Q104:** A **Q105:** C **Q106:** C **Q107:** C **Q108:** B **Q109:** B **Q110:** C
Q111: C **Q112:** C **Q113:** B **Q114:** A **Q115:** C **Q116:** B **Q117:** A **Q118:** A **Q119:** B **Q120:** B
Q121: C **Q122:** B